FIRE VOLT

Project Technical Manual

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Group Project — Fire Volt

Version 1.0

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1. Project Overview

Fire Volt is an innovative engineering project that combines three cutting-edge subsystems into a single autonomous car chassis. The goal is to demonstrate sustainable energy recovery and environmental responsibility — all within a compact, self-contained vehicle platform. WE ARE HERE MAKING IT'S PROTOTYPE!

The car is powered by rechargeable batteries and controlled wirelessly via an ESP-32 microcontroller. A combustion chamber mounted on the chassis generates heat, which is harvested by Thermoelectric Generator (TEG) modules and converted into usable electrical energy through two custom-designed boost converters. This energy is fed back into the battery pack, extending operating time.

Additionally, the exhaust smoke from the combustion chamber is routed through a filtration system. A relay-controlled fan draws the smoke into a filtration chamber where carbon soot is captured and processed into usable ink — turning waste into a valuable byproduct which is the ink generated.

Project Goals

Subsystem	Purpose
Car Motion	Autonomous movement via ESP-32 + motor driver + 4 DC motors powered by rechargeable batteries
Heat to Electricity	TEG modules capture combustion heat; custom boost converters step up voltage to recharge batteries
Filtration	Relay-controlled fan draws exhaust smoke into a chamber; carbon soot is converted into ink

2. System Architecture

The Fire Volt system is built around a central car chassis that serves as the mounting platform for all three subsystems. The rechargeable battery pack is the central power node — it provides power to the ESP-32, the motor driver, and the relay. The TEG + boost converter chain feeds energy back into this battery pack, creating a partial energy recovery loop.

The ESP-32 acts as the brain, issuing PWM motor control signals and digital relay control signals. The combustion chamber plays a dual role: it is the heat source for the TEG modules and also the origin of smoke for the filtration system.

3. Component Reference

This section documents every component used in the Fire Volt project, including its specifications, working principle, and role in the system.

3.1 ESP-32 Microcontroller

The ESP-32 is a low-cost, low-power system-on-chip microcontroller with integrated Wi-Fi and Bluetooth. It is the central brain of the Fire Volt car, responsible for processing control signals and driving the motor driver and relay.

Specification	Details
Operating Voltage	3.3V (logic) / 5V via USB
CPU	Dual-core Xtensa LX6, up to 240 MHz
GPIO Pins	34 programmable GPIO pins
Connectivity	Wi-Fi 802.11 b/g/n, Bluetooth 4.2 / BLE
PWM Output	Up to 16 PWM channels
Role in project	Sends PWM signals to motor driver; sends GPIO signal to relay module

In Fire Volt, the ESP-32 receives remote control commands and translates them into PWM signals for the motor driver. It also controls the filtration fan by toggling the relay through a GPIO pin.

3.2 Motor Driver (L298N)

The L298N is a dual full-bridge H-bridge motor driver capable of driving two DC motors independently with direction and speed control. In Fire Volt it drives all four DC motors (paired).

Specification	Details
Input Voltage	5V-35V (motor supply); 5V (logic)
Max Current	2A per channel (4A peak)
Control Inputs	PWM on ENA/ENB; IN1-IN4 for direction
Role in project	Drives all 4 DC motors — speed via PWM and direction via IN pins

The L298N receives PWM signals from the ESP-32 and adjusts the power delivered to each motor pair, enabling forward, reverse, and turning manoeuvres.

3.3 DC Motors (x4)

Four standard DC geared motors are used — one per wheel. These convert electrical energy from the motor driver into rotational mechanical energy to drive the chassis.

- **Typical Voltage:** 3V-12V DC
- **Speed:** 100-300 RPM (geared variants)
- **Mounting:** Directly attached to each wheel axle on the chassis
- **Role in project:** Four-wheel drive — all four motors run for forward/reverse; differential speeds for turns

3.4 Rechargeable Battery Pack

The battery pack is the primary energy reservoir for the entire system. It powers the ESP-32, motor driver, and relay, and is also recharged by the TEG + boost converter chain during operation.

- **Chemistry:** Li-ion or Li-Po (recommended: 2S 7.4V pack)
- **Capacity:** 2000-5000 mAh depending on desired runtime
- **Discharge Rate:** Minimum 10C for peak motor current draw
- **Role in project:** Central power supply; receives harvested energy from boost converter 2

3.5 TEG Modules (Thermoelectric Generators)

A TEG module converts a temperature difference directly into electrical voltage using the Seebeck effect. When one face is hot (attached to the combustion chamber) and the other is cold, a DC voltage is generated. In Fire Volt, 12 to 16 TEG modules are connected in series to produce a usable combined voltage.

Specification	Details
Module used	TEC1-12706 (or equivalent)
Voltage per module	~0.5V - 1V depending on delta-T
Series count	12 to 16 modules
Total output voltage	Approx. 8V - 16V (before boost)
Physical mounting	Hot side bonded to combustion chamber wall; cold side exposed to air or heatsink
Role in project	Converts combustion heat into DC voltage fed into boost converter 1

The series connection is critical — individual TEG modules produce very low voltage. Series stacking multiplies the voltage while keeping current constant, giving a workable input to boost converter 1.

3.6 Boost Converters (Custom-Designed)

A boost converter is a DC-DC power converter that takes a lower input voltage and outputs a higher voltage. Fire Volt uses two custom-designed boost converters in cascade to achieve the final battery-compatible charging voltage.

- **Boost Converter 1:** Takes TEG series output (~8-16V); first step-up stage; designed by the Fire Volt team
- **Boost Converter 2:** Takes output of BC1; second step-up stage; outputs stable voltage for battery charging
- **Key components:** Inductor, MOSFET switch, diode, capacitor, PWM controller IC
- **Role in project:** Together they convert low/variable TEG voltage to a regulated charging voltage for the battery pack

3.7 Relay Module

A relay is an electrically operated switch. A low-power GPIO signal from the ESP-32 energises the relay coil, which closes a high-power contact to switch the fan on or off, allowing safe control of higher-voltage loads.

- **Coil voltage:** 5V DC
- **Switching capacity:** Up to 10A / 250V AC or 30V DC
- **Protection:** Flyback diode (1N4007) across coil to protect ESP-32 GPIO

- **Role in project:** Switches the inbuilt fan ON/OFF on command from the ESP-32

3.8 Inbuilt Fan

A small DC fan is integrated into the filtration subsystem. When activated via the relay, it creates suction to draw exhaust smoke from the combustion chamber pipe into the filtration chamber.

- **Type:** Small brushless DC fan (e.g. 12V 40mm or 60mm)
- **Control:** Switched via relay module; ON/OFF
- **Placement:** Inlet of the filtration chamber
- **Role in project:** Actively draws smoke from combustion exhaust into the filtration chamber

3.9 Filtration Chamber

The filtration chamber is a custom-built enclosure that receives smoke from the exhaust pipe. Carbon soot particles are captured and processed with a binding agent to produce usable ink — turning waste into a resource.

- **Input:** Smoke from combustion chamber via exhaust pipe
- **Output 1:** Cleaned exhaust air expelled to atmosphere
- **Output 2:** Captured carbon soot processed into ink
- **Role in project:** Environmental control component — prevents soot release; produces ink as byproduct

3.10 Combustion Chamber

The combustion chamber is the central heat source of the Fire Volt system. It burns fuel to produce heat harvested by the TEG modules, and simultaneously generates exhaust smoke piped to the filtration subsystem.

- **Heat output:** High enough to create meaningful delta-T across TEG modules
- **TEG mounting:** TEG modules bonded directly to outer wall surface
- **Exhaust outlet:** Pipe leading to filtration fan and chamber
- **Safety:** Must be thermally insulated from other electronics on the chassis
- **Role in project:** Dual role — heat source for energy harvesting AND smoke source for filtration

4. How It All Works Together

This section describes the integrated operation of Fire Volt — how the three subsystems interact with each other as a unified system when the car is running.

4.1 Car Motion Subsystem

At power-on, the rechargeable battery pack supplies voltage to the ESP-32 (via a 3.3V regulator) and to the motor driver (direct). The ESP-32 boots and listens for control inputs via Wi-Fi or Bluetooth.

When a movement command is received, the ESP-32 sets the appropriate IN1-IN4 direction pins on the L298N and applies a PWM duty cycle to the ENA/ENB enable pins. This varies the average voltage delivered to each motor pair, controlling both speed and direction. To turn, the left and right motor pairs receive different speeds (differential drive). The four DC motors then drive all four wheels accordingly.

4.2 Heat-to-Electricity Subsystem

While the car is running, the combustion chamber is active, burning fuel to generate heat. The 12-16 TEG modules bonded to the chamber wall generate a small DC voltage each via the Seebeck effect. Because the modules are wired in series, their voltages add up to approximately 8V-16V combined.

This combined voltage feeds into Boost Converter 1, then into Boost Converter 2 (both custom-built by the Fire Volt team), which outputs a final regulated voltage suitable for charging the Li-ion battery pack. This creates a partial closed-loop energy cycle — combustion heat that would otherwise be wasted is recovered to extend battery runtime.

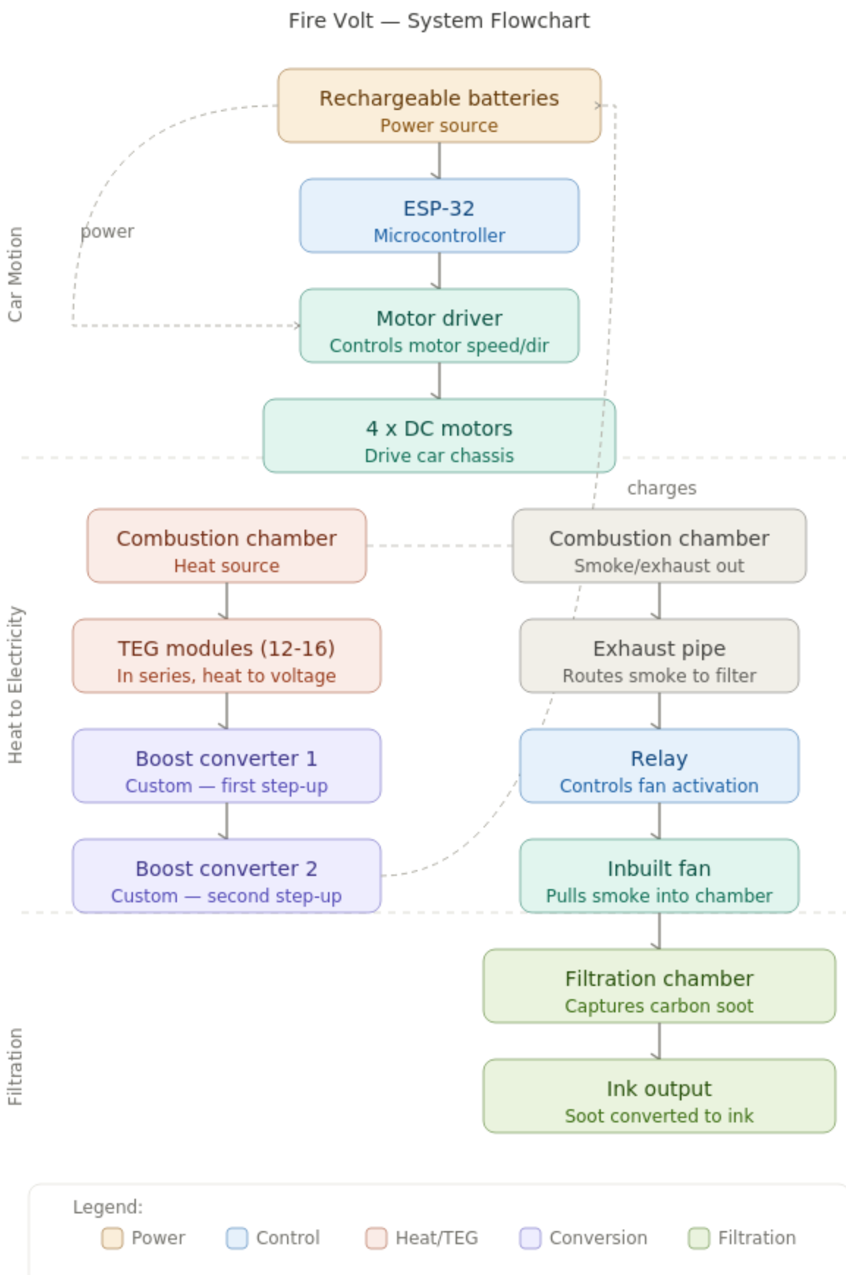
4.3 Filtration Subsystem

The combustion chamber also produces exhaust smoke. The ESP-32 sends a GPIO signal to the relay module, which closes its contact and powers the inbuilt DC fan. The fan creates suction that draws the smoke into the filtration chamber, where carbon soot particles are captured on the filter medium.

Periodically, the accumulated carbon soot is collected and mixed with a suitable binding agent (e.g. gum arabic) to form usable ink — turning an environmental waste product into a functional material. The relay gives the ESP-32 complete control over when the fan runs; in practice it is activated whenever the combustion chamber is operating.

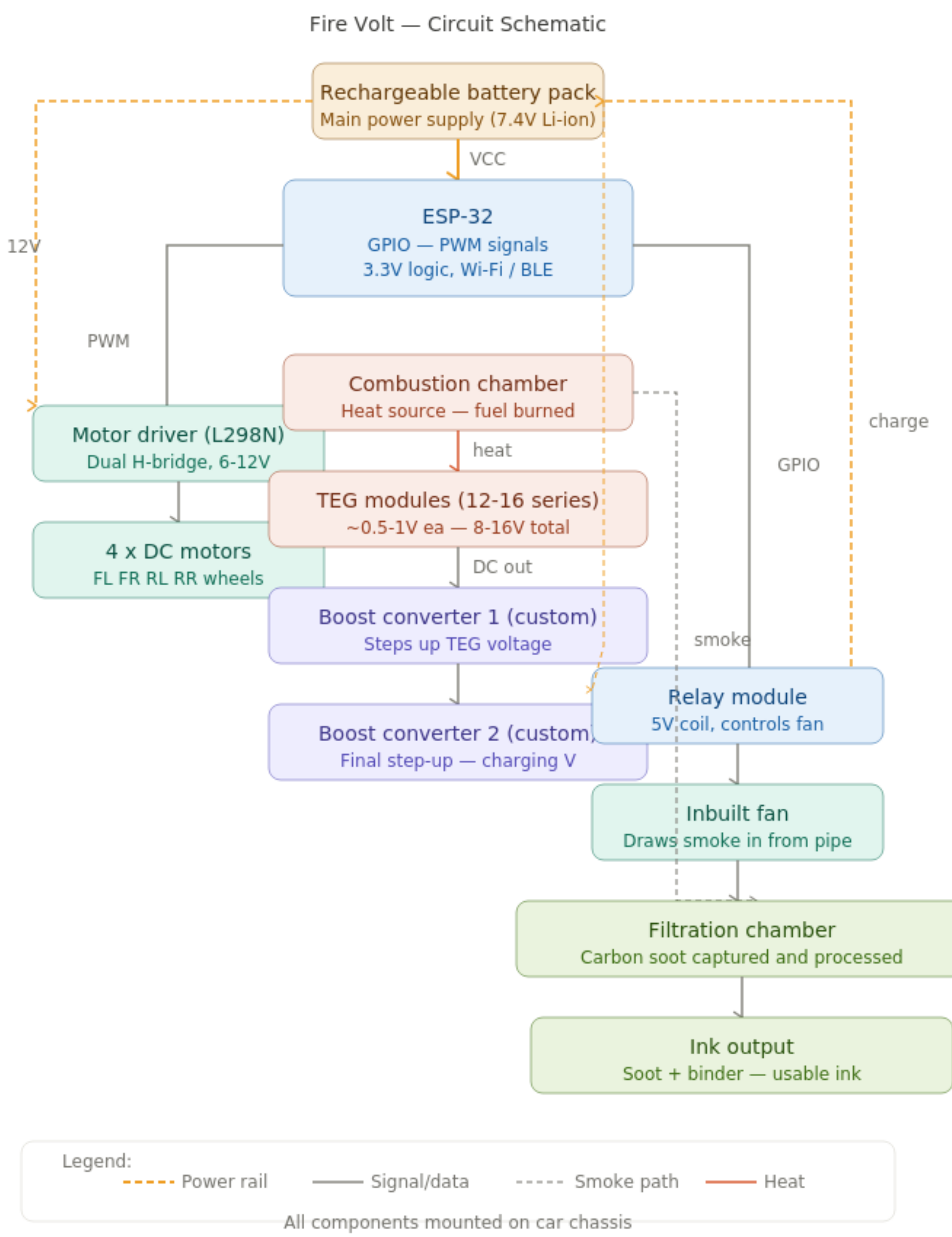
5. System Flowchart

The flowchart below illustrates the complete operational flow of Fire Volt across all three subsystems. Each subsystem is colour-coded: Car Motion (blue/teal), Heat-to-Electricity (coral/purple), and Filtration (green). Dashed arrows show the energy recovery feedback loop from the boost converters back to the battery pack.



6. Circuit Diagram

The circuit schematic below shows all electrical connections between components. Orange dashed lines are power rails from the battery pack. Solid grey lines are control signals from the ESP-32. The red line shows the heat path from TEG modules to boost converters and back to the battery as charging current.



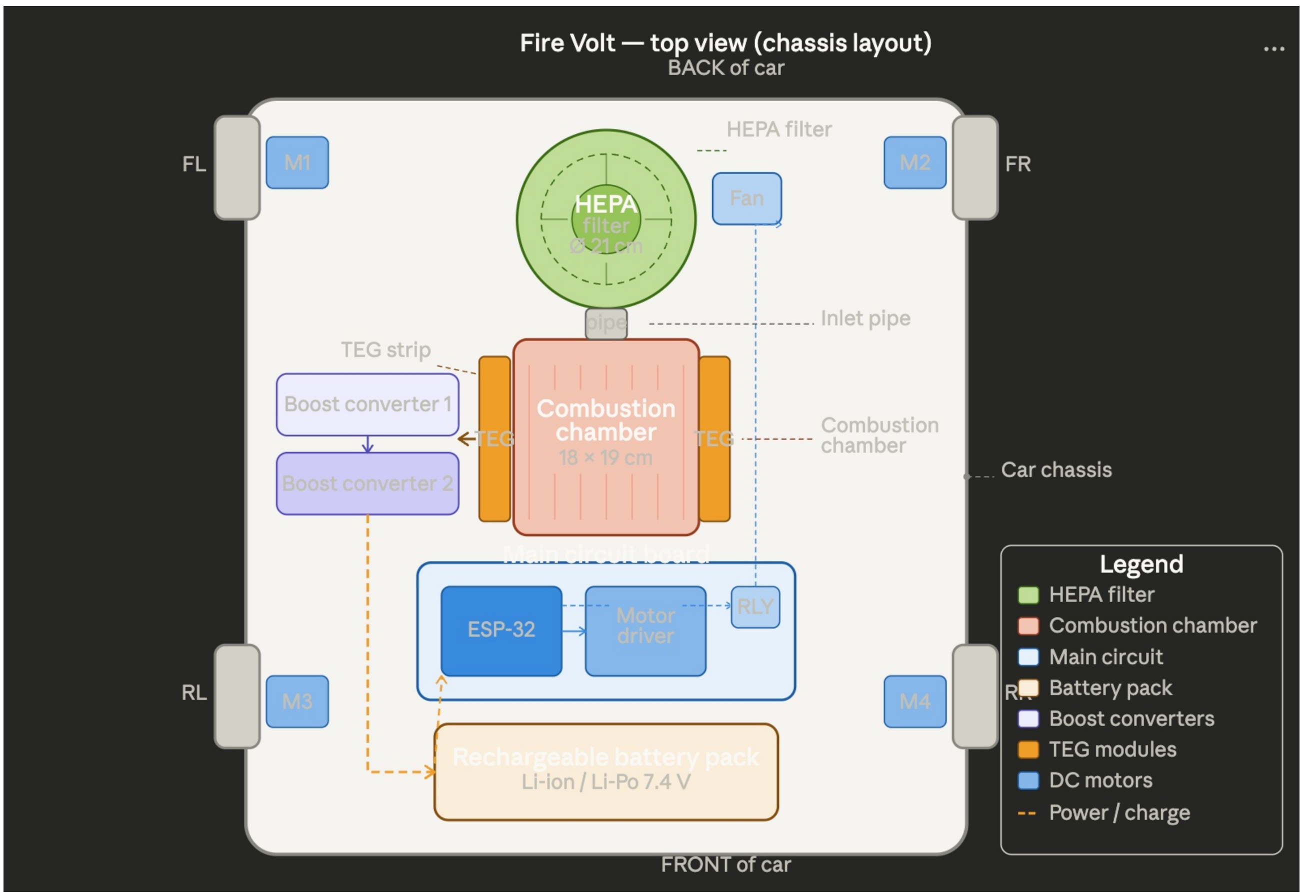
KEY NOTES:

- **Power rail:** Battery (+) connects to: ESP-32 VIN, Motor Driver VS, Relay VCC. Battery (-) is common ground.
- **ESP-32 regulation:** Use an AMS1117 3.3V regulator between battery and ESP-32 VCC if not powering via USB.
- **Relay protection:** A flyback diode (1N4007) across the relay coil is mandatory to protect ESP-32 GPIO from inductive kickback.
- **TEG polarity:** Ensure consistent polarity when wiring TEG modules in series — reversing one module subtracts voltage.
- **Boost converter output:** Verify output voltage matches battery charging spec before connecting — overcharging Li-ion is a fire hazard.

7. Chassis Layout Diagrams

The diagrams below show how all Fire Volt components are physically arranged on the car chassis, as seen from the top and side. The HEPA filter (Ø 21 cm) is positioned at the back, immediately followed by the combustion chamber (18 × 19 cm) with TEG modules on its sides. The inlet pipe connects the combustion chamber exhaust directly into the base of the HEPA filter. The main circuit board and battery pack occupy the forward section of the chassis.

Top View — Chassis Layout



Circuit Schematic